

Integrals – 5

The gamma and beta functions and Wallis' formulas

Homework

1 сентября 2026 года

Theoretical minimum. The gamma and beta functions and the relation between them; the reflection and duplication formulas; Wallis' formulas and Wallis' product; Stirling's asymptotics; the substitutions $x = \sin^2 t$ and $x \mapsto \frac{x}{1+x}$; the universal template $\int_0^\infty x^{s-1} e^{-ax^b} dx$.

1. Prove that $\int_0^\infty x^{s-1} e^{-ax^b} dx = \frac{1}{b} a^{-s/b} \Gamma(s/b)$ for $a, b, s > 0$, and deduce the values of $\int_0^\infty e^{-x^3} dx$, $\int_0^\infty x^5 e^{-x^4} dx$ and $\int_0^\infty x^3 e^{-x^2} dx$.
2. Prove Legendre's duplication formula using the beta integral and the substitution $x = \sin^2 t$.
3. Compute $\int_0^1 \frac{dx}{\sqrt[n]{1-x^n}}$ through the beta function and find the limit of the answer as $n \rightarrow \infty$.
4. Prove Wallis' product $\frac{\pi}{2} = \prod \frac{(2n)^2}{(2n-1)(2n+1)}$ from Wallis' formulas and deduce Stirling's formula up to a constant; determine the constant $\sqrt{2\pi}$ separately.
5. Compute $\int_0^{\pi/2} \tan^a x dx$ for $|a| < 1$ and explain why the answer does not depend on the sign of a .
6. Compute $\int_0^1 \frac{dx}{\sqrt{-\ln x}}$ and $\int_0^1 \left(\ln \frac{1}{x}\right)^{s-1} dx$.
7. Prove that Γ is logarithmically convex and state the Bohr–Mollerup theorem: it is the unique function with $\Gamma(1) = 1$, $\Gamma(s+1) = s\Gamma(s)$ and logarithmic convexity.
8. (Integration Bee) Reduce to the gamma function and compute $\int_0^\infty \frac{dx}{e^x+1}$ (2020), $\int_0^\infty \frac{x dx}{e^x-1}$ and $\int_{-\infty}^\infty \frac{e^{-x^2}}{1+e^{2x}} dx$ (2026); in the last one it is symmetry rather than the gamma function that helps – explain why.

9. Compute the volume of the n -dimensional ball through the gamma function, check the answer for $n = 2, 3, 4$ and find the asymptotics of the volume as $n \rightarrow \infty$.
10. Prove that $\int_0^\infty \frac{\sin x}{x^s} dx = \frac{\pi}{2\Gamma(s) \sin \frac{\pi s}{2}}$ for $0 < s < 2$, and recover the classical value at $s = 1$.